

STA305H5S Experimental Design Project

Winter 2026

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Section 1, Option A: fertdat

The **fertdat** dataset follows a completely randomized design with 12 independent plots as experimental units. Each plot was randomly assigned to one of two treatments, fertilizer A or fertilizer B, with 6 plots per treatment. See that independence between plots is assumed based on the random assignment. The research question we will investigate is whether there is a difference on the mean yield of wheat between fertilizer A and fertilizer B at significance level $\alpha = 0.05$.

The experimental units are a singular plot of wheat, the response variable is the yield of wheat from an individual plot and the treatment factor is the type of fertilizer used on the plot of wheat.

Table 1 below shows the summary statistics of the **fertdat** dataset while Figure 1 shows the yield of wheat by fertilizer type.

Table 1: Summary statistics of yield by fertilizer type

trt	n	mean	sd	median	min	max
A	6	22.18	3.86	22.4	16.5	26.9
B	6	20.48	7.00	21.1	11.4	28.5

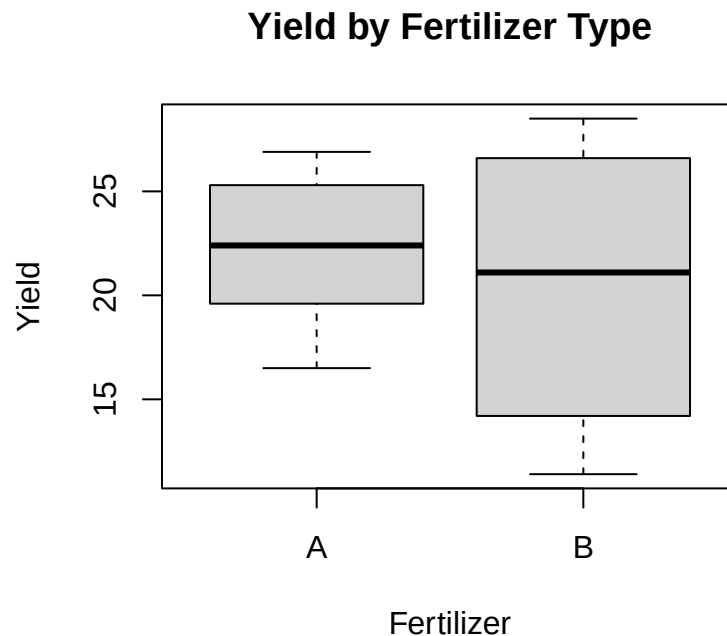


Figure 1: Boxplot of wheat yields by fertilizer type

From Figure 1, we observe that the centers of the two groups appear similar, although fertilizer B shows greater variability in yields compared to fertilizer A. This implies that the variances of treatment A and treatment B are unequal, however, to ensure this assumption, we will still test the variance later. For now, we proceed to check whether the data can be assumed to be approximately normal.

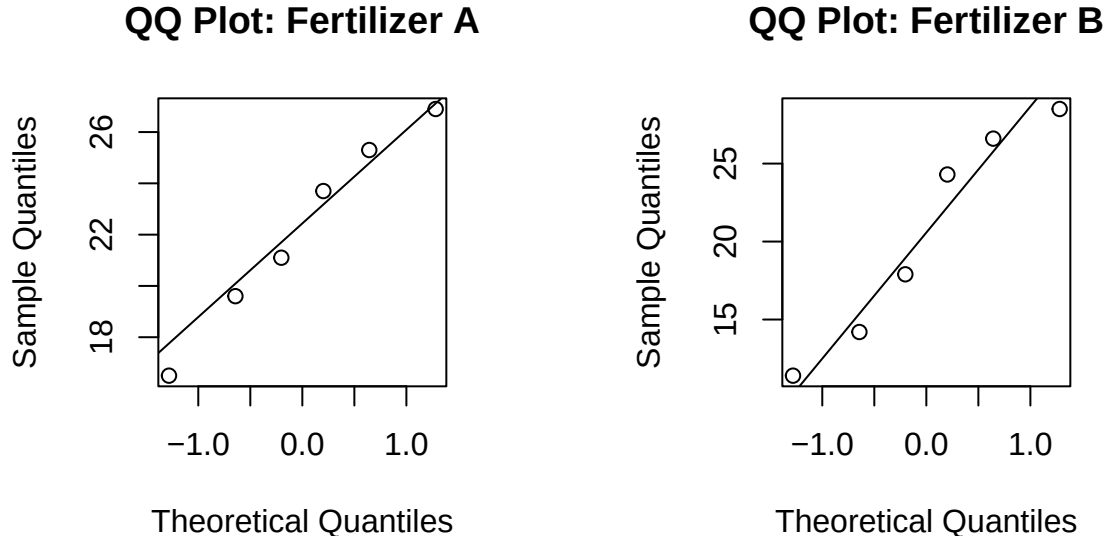


Figure 2: QQ plots of yield by fertilizer type

In figure 2, residuals from the two-sample model were examined. The QQ plots show no strong deviations from normality, and there are no clear outliers. Thus, the data does not show strong evidence against the assumption of normality.

Although the data indicates evidence for a dataset with unequal variances from Figure 1 and valid normality assumption from Figure 2, we will still directly test the assumption of equal variance as noted previously.

Thus, an F-test was conducted to compare the variances of yield between fertilizer A and fertilizer B (see Appendix C). The test returned not statistically significant, where $F(5, 5) = 0.304$ with $p\text{-value} = 0.217$. Since the $p\text{-value}$ is greater than the significance level 0.05, we fail to reject the hypothesis that the variances of yield between fertilizer A and fertilizer B of wheat is equal. So, there is not enough evidence against the fact that variances differ between the two treatments.

Nevertheless, due to small sample size and the unequal spread of data that Figure 1 showed, we will still proceed with Welch's two-sample t-test as a precaution against unequal variances. We will test the following hypothesis:

$$H_0 : \mu_A = \mu_B \quad \text{vs.} \quad H_a : \mu_A \neq \mu_B$$

Note that we do not use a randomization test as the sample sizes are equal in both treatments and Figure 2 shows reasonable evidence for assuming normality.

Through conducting Welch's two-sample t-test (see Appendix C), the t value is 0.5211 at 8 degrees of freedom. The resulting $p\text{-value} = 0.6168$. Since the $p\text{-value} = 0.6168 > \alpha = 0.05$, and since the confidence interval calculated from the two-sample t-test is $(-5.86, 9.26)$ which contains 0, we fail to reject the null hypothesis at the 5% significance level.

Hence, there is insufficient evidence to suggest that fertilizer A and fertilizer B produce different mean yields of wheat at the 5% level. That is, the data does not provide evidence that one fertilizer leads to higher yield than the other.

Section 2

Similar to Section 1, the future study will use a completely randomized design with individual, independent wheat plots as experimental units. Each plot of wheat will be randomly assigned to either treatment, fertilizer A or fertilizer B. On the other hand, the response variable is still wheat yield. Furthermore, the effect size is defined as $\mu_A - \mu_B$, which is the mean of all yields of wheat plots using fertilizer A minus the mean of all yields of wheat plots using fertilizer B. Estimating using the **fertdat** dataset from Section 1, this value is 1.7 (see Appendix C). While 1.7 is estimated from the current observed data, it represents a small difference relative to the overall variability, however, detecting such a difference is practically meaningful, so this value is a meaningful effect size for the **fertdat** dataset.

Next, we would like to examine how many plots for each type of fertilizer we would need to allocate to achieve 80% power at significance level $\alpha = 0.05$. A two-sided, two-sample t-test on power was conducted to detect $\mu_A - \mu_B = 1.7$ (see Appendix C), and also by using a population standard deviation as an approximation. Note that the standard deviation is estimated using all observations to reflect overall variability in yield across treatments.

As a result, the required sample size is 163 plots per treatment group. This means a total of 326 plots are needed to achieve 80% power to detect a true mean difference of 1.7 at the 5% significance level.

Then, to ensure that $n = 163$ is the correct number of samples to achieve 80% power, we will check against a simulation-based calculation that simulates the dataset but with each treatment group having 163 data points instead. Note that the data is simulated under the assumption that the true mean difference between the two treatment groups is 1.7, and that both treatments have the same standard deviation as previously estimated.

Sure enough, the simulation repeatedly tests two groups of size 163 under the assumed effect size and variability. Power is estimated as the proportion of tests with p-values below 0.05. The result of 0.802 confirms that the power t-test was correct.

Hypothetically, if the true effect size is smaller than 1.7, the required sample size increases because smaller differences are harder to detect. On the other hand, if variability is larger, the standard error increases, which also reduces power and requires more samples. Understandably, both of these factors directly raise the amount of samples needed to maintain 80% power.

Section 3A, Option A: painstudy

This dataset follows a completely randomized design where each experimental unit is independently assigned to one of three treatments A, B, or C. The experimental units in the **painstudy** data set is the individual subject to the type of treatment. The response variable is the level of pain they experience from the type of treatment, while the treatments are the three types of methods used on the experimental units, A, B, or C. We will investigate whether the mean pain levels differ across the three treatments. This will be accomplished by fitting a one-way ANOVA to the **painstudy** dataset. We test the hypothesis:

$$H_0 : \mu_A = \mu_B = \mu_C \quad \text{vs.} \quad H_a : \text{at least one mean differs}$$

The ANOVA table is as follows in Table 2.

Table 2: Summary of the one-way ANOVA for the painstudy dataset

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
trt	2	355.84	177.920	98.919	0
Residuals	147	264.40	1.799	NA	NA

For the ANOVA model to be valid, it requires two significant assumptions. Figure 3 below details the ANOVA table assumptions. That is, the residuals are approximately normally distributed normality in the dataset, and that the variances are spread equally across treatment groups.

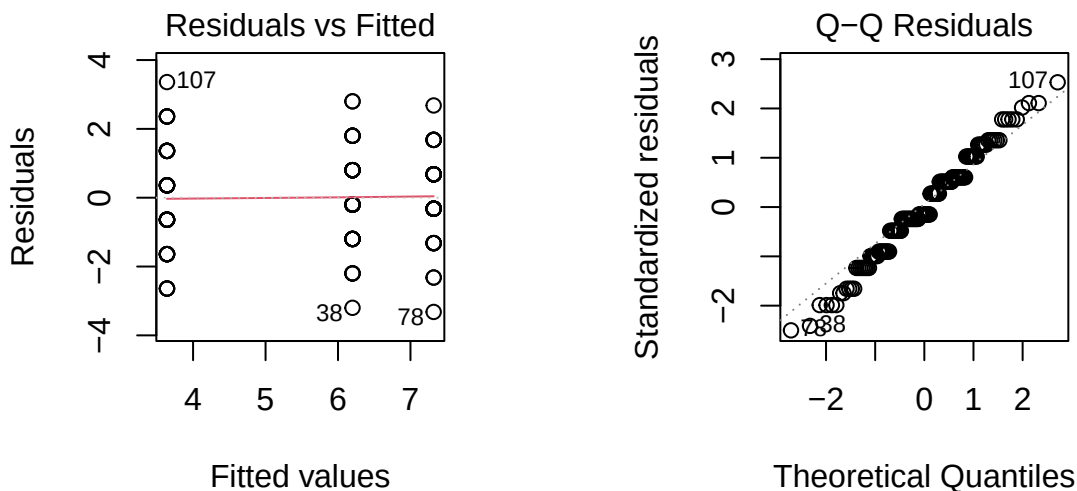


Figure 3: Plots to check normality of residuals in data and homogeneity of variances.

From Figure 3, the residuals versus fitted plot shows no clear pattern, which supports the assumption of constant variance. Furthermore, the Q-Q plot shows points close to the line without outliers, which supports the normality assumption. Also, the F-test from Table 2 yields a very small p-value, which is below 0.05. So, we reject the null hypothesis and conclude that there is evidence that suggests at least one treatment mean differs. These results indicate that the ANOVA assumptions are reasonable and we proceed to post-hoc comparisons.

By conducting Tukey's post-hoc comparisons (see Appendix C), the result of all pairwise differences is statistically significant ($p < 0.05$). The estimated differences indicate that treatment B has the highest mean pain level, followed by treatment A, while treatment C has the lowest mean pain level.

Thus, the ordering of treatments in terms of mean pain is $B > A > C$. We conclude that treatment C appears to be the most effective at reducing pain while treatment B is the least effective treatment at reducing pain.

Section 3B

We investigate whether the wear of shoes on the left and right foot of a boy is the same or different using the **shoedat_obs** dataset. To achieve this, we leverage the randomized blocking design that the dataset uses. The experimental units are the two shoes worn by each boy while the treatments themselves are the shoe type, labelled A and B, randomly assigned to each foot within a boy. The blocking variable is the variable **boy**, while the response variable is the wear of the shoe dependent on which foot it is on.

Understand that blocking is appropriate in this dataset since each variable **boy** is subject to both treatments. Due to the nature of this dataset, the treatment factor is the type of shoe (sideA or sideB), which are subject to many factors outside of the researcher's control. For example, we cannot control how the boy wearing each treatment lives their life. Each experimental unit in this case could have different activity levels, walking postures, and sentiments to their shoes (how they treat what they wear). Thus, due to this vast variability, blocking is necessary to control the amount of noise in the dataset.

We fit the **shoedat_obs** dataset to an additive model seen in Table 3:

Table 3: Additive model for the shoedat obs dataset

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
trt	1	9.092	9.092	1.74	0.220
factor(boy)	9	20.214	2.246	0.43	0.888
Residuals	9	47.036	5.226	NA	NA

In table 3, the treatment p-value is $0.22 > 0.05$. Therefore, we fail to reject the null hypothesis and conclude that there is insufficient evidence to suggest a difference in mean wear between shoe A and shoe B in each boy. Similarly, the block factor has a large p-value = 0.888, suggesting that there is no strong evidence of differences in average wear across boys. That is, the block effect is not statistically significant. However, blocking is still appropriate because it accounts for pairing within each boy. Even when the block effect is not significant, the design ensures that comparisons between treatments are made within the same subject, which improves precision.

In comparison, an unblocked analysis would ignore the pairings of shoes to boys and treat all observations as independent. This would increase the error variance by mixing between-boy variability with noise. What the randomized block design does is separate this variability and focuses the comparison within each boy, leading to a more precise estimate of the treatment effect.

Section 3C

This study aims to investigate whether the mean value differs from applying 4 treatments, 4 dogs and 4 weeks, utilizing the fact that the **BR_LatSq** is a Latin square design. The experimental units are the individual observations of response for each dog in each week. Each treatment appears exactly once in every row and column. Note that treatments in the Latin square design are assigned to balance both blocking factors, **Dogs** in the rows, and ****Weeks** in the columns, to control for differences between dogs and changes in weeks. The response variable is the value given a type of dog in a specific week given a certain treatment.

The results of the Latin square model is seen as Table 4 below, summarized in the form of an ANOVA table.

Table 4: Latin square ANOVA for the BR LatSq dataset

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Dogs	3	2.832	0.944	1.793	0.248
Weeks	3	0.547	0.182	0.346	0.794
Treat	3	4.757	1.586	3.012	0.116
Residuals	6	3.159	0.526	NA	NA

We test the hypotheses:

$$H_0 : \mu_A = \mu_B = \mu_C = \mu_D \quad \text{vs.} \quad H_a : \text{at least one treatment mean differs}$$

From Table 4, The treatment effect has a p value of 0.0485, which is below 0.05. Therefore, we reject the null hypothesis and conclude that there is evidence that suggests at least one treatment mean differs after accounting for differences between **Dogs** and ****Weeks****. As a result, this indicates that treatment type affects the response beyond variability due to subjects and time.

Section 4

The **wtlossdat** dataset is a full 2^3 factorial design with three factors: A, B, and C. The experimental units are the individual runs of the process that produces the response y. Each factor has two levels which are the values they take. In this case, each factor is coded as either -1, which represents low or 1, which represents high. Finally, the response variable is y.

Table 5 below denotes the main effects and the interaction effects of the three factors in the 2^3 factorial design.

Table 5: Estimated main and interaction effects for factors A, B, and C

Effect	Estimate
A	0.525
B	-1.975
C	0.575
A $\times$ B	0.175
A $\times$ C	0.625
B $\times$ C	-0.575
A $\times$ B $\times$ C	0.175

From the table, factor B has the largest main effect at -1.975, indicating a strong negative effect on the response. This suggests that setting B from -1 to +1 substantially decreases the response.

Factors A at 0.525 and C at 0.575 have medium positive effects, meaning that increasing the level of these factors increases the response.

Among the interaction effects in Table 6, A:C at 0.625 and B:C at -0.575 stand out as being larger effects than the other two interactions. As a result, this suggests a potential interaction between A and C and between B and C while the other two interaction effects are smaller in magnitude and are likely negligible in interaction.

We fit a linear model in the table below that shows the relationship of all main and interaction effects.

Table 6: Linear model coefficients for main and interaction effects

Term	Estimate	Std..Error	t.value	Pr...t..
(Intercept)	0.637	NaN	NaN	NaN
A	0.262	NaN	NaN	NaN
B	-0.987	NaN	NaN	NaN
C	0.288	NaN	NaN	NaN
A $\times$ B	0.087	NaN	NaN	NaN
A $\times$ C	0.312	NaN	NaN	NaN
B $\times$ C	-0.288	NaN	NaN	NaN
A $\times$ B $\times$ C	0.088	NaN	NaN	NaN

Since the model includes all main effects and interactions, this results in a saturated model with no residual degrees of freedom. This prevents further assessment of model assumptions and tests. We proceed to make inferences by using the main and interaction plots of the effects.

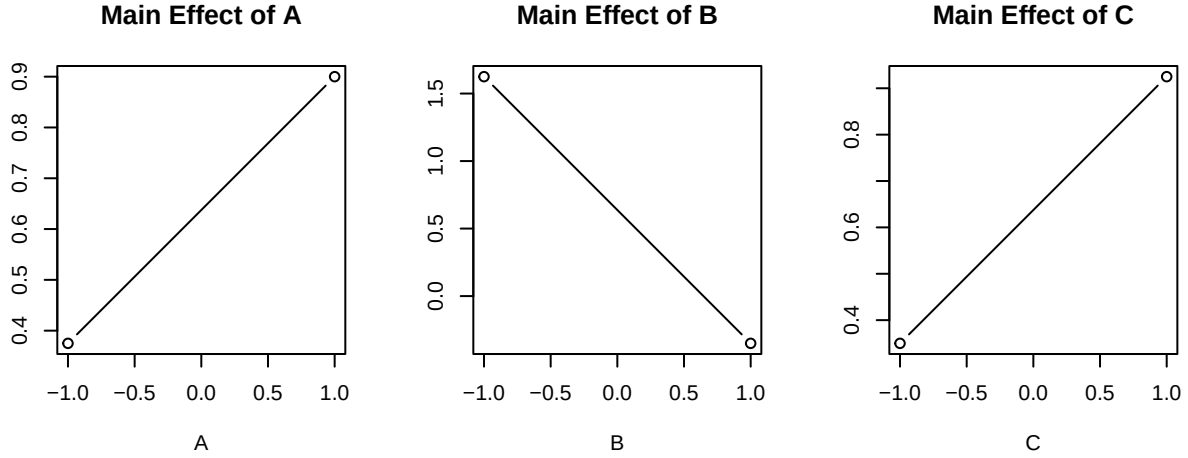


Figure 4: Plots detailing main effects of the three factors

From Figure 4, the plots show that factor B has the largest effect and is negative, which means increasing B from low to high reduces the response. Factors A and C have smaller positive effects, which means higher levels increase the response, yet at a smaller degree. Notably, these visual results accurately reflect the values of the main effects that Table 5 provides.

Now below, we observe Figure 5 which shows the plots of the two-way interactions that occur between the three factors.

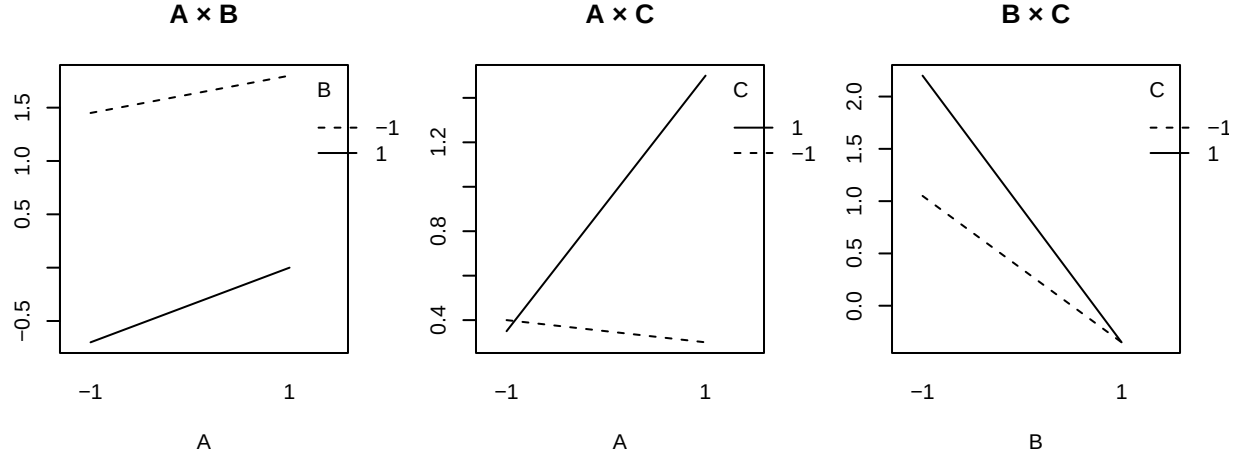


Figure 5: Plots showing the two-way interactions between the three factors

Figure 5 shows evidence of interaction, particularly the largest interactions are $A \times C$ and $B \times C$. Observe that all interaction plots show non parallel lines, which indicates interaction. However, all these effects are smaller than the main effect of B as seen in Table 6. That is, factor B still has the greatest impact on the response variable, with some contribution from the interactions of the factors.

Based on the estimated effects and the plots, the combination of factor settings that maximizes the response is $A = +1$, $B = -1$, and $C = +1$. This combination produces the highest observed response and is consistent with the effect estimates in Table 5, where B should be low and A and C should be high.

Appendix

This appendix contains all R code, R code calculations, and supporting outputs referenced in the main report.

Appendix A

Section 1: R code for the boxplot graph that displays yield of wheats by treatment in Figure 1.

```
par(mfrow = c(1,1), pin = c(3, 2))

boxplot(fert ~ trt,
        data = fertdat,
        main = "Yield by Fertilizer Type",
        xlab = "Fertilizer",
        ylab = "Yield")
```

Section 1: R code of a Q-Q plot that displays the normality of the **fertdat** dataset in Figure 2.

```
par(mfrow = c(1,2), pin = c(1.5, 1.5))

qqnorm(fertdat$fert[fertdat$trt == "A"],
        main = "QQ Plot: Fertilizer A")
qqline(fertdat$fert[fertdat$trt == "A"])

qqnorm(fertdat$fert[fertdat$trt == "B"],
        main = "QQ Plot: Fertilizer B")
qqline(fertdat$fert[fertdat$trt == "B"])
```

Section 3A: R code of plots that check homogeneity of variances and normality in Figure 3

```
par(mfrow = c(1,2), pin = c(1.5, 1.5))
plot(anova_model, which = 1)
plot(anova_model, which = 2)
```

Section 4: R code of plots that visualize the main effects of factors A, B, and C in Figure 4.

```
par(mfrow = c(1,3), pin = c(1.5, 1.5))

with(wtlossdat, {
  plot(c(-1,1), tapply(y, A, mean), type = "b",
        xlab = "A", ylab = "Mean y", main = "Main Effect of A")

  plot(c(-1,1), tapply(y, B, mean), type = "b",
        xlab = "B", ylab = "Mean y", main = "Main Effect of B")

  plot(c(-1,1), tapply(y, C, mean), type = "b",
        xlab = "C", ylab = "Mean y", main = "Main Effect of C")
})
```

Section 4: R code of plots that visualize the two-way interactions between the factors A, B, and C in Figure 5.

```
par(mfrow = c(1,3), pin = c(1.5, 1.5))

interaction.plot(wtlossdat$A, wtlossdat$B, wtlossdat$y,
  main = "A × B", xlab = "A", trace.label = "B")

interaction.plot(wtlossdat$A, wtlossdat$C, wtlossdat$y,
  main = "A × C", xlab = "A", trace.label = "C")

interaction.plot(wtlossdat$B, wtlossdat$C, wtlossdat$y,
  main = "B × C", xlab = "B", trace.label = "C")
```

Appendix B

Section 1: R code for the summary statistics of **fertdat** in a table using kable for Table 1.

```
library(dplyr)
library(knitr)
library(ggplot2)
library(scidesignR)
library(kableExtra)

summary_table <- fertdat %>%
  group_by(trt) %>%
  summarise(
    n = n(),
    mean = mean(fert),
    sd = sd(fert),
    median = median(fert),
    min = min(fert),
    max = max(fert)
  )

kable(summary_table,
  format = "latex",
  booktabs = TRUE,
  caption = "Summary statistics of yield by fertilizer type",
  digits = 2) %>%
  kable_styling(latex_options = "hold_position")
```

Section 3A: R code that constructs an ANOVA table for the **painstudy** dataset in Table 2.

```
library(dplyr)
library(knitr)
library(kableExtra)
```

```

anova_model <- aov(pain ~ trt, data = painstudy)
anova_table <- summary(anova_model)[[1]]
anova_table <- cbind(Source = rownames(anova_table), anova_table)
rownames(anova_table) <- NULL

kable(anova_table,
      format = "latex",
      booktabs = TRUE,
      caption = "Summary of the one-way ANOVA for the painstudy dataset",
      digits = 3) %>%
  kable_styling(latex_options = "hold_position")

```

Section 3B: R code that constructs an additive model in the form of an ANOVA table for Table 3.

```

library(knitr)
library(kableExtra)

shoedat_long <- data.frame(
  boy = rep(shoedat_obs$boy, 2),
  trt = factor(rep(c("A", "B"), each = nrow(shoedat_obs))),
  wear = c(shoedat_obs$wearA, shoedat_obs$wearB)
)

model <- aov(wear ~ trt + factor(boy), data = shoedat_long)

anova_table <- summary(model)[[1]]
anova_table <- cbind(Source = rownames(anova_table), anova_table)
rownames(anova_table) <- NULL

kable(anova_table,
      format = "latex",
      booktabs = TRUE,
      caption = "Additive model for the shoedat obs dataset",
      digits = 3) %>%
  kable_styling(latex_options = "hold_position")

```

Section 3C: R code that constructs a Latin Square version of an ANOVA table in Table 4.

```

library(knitr)
library(kableExtra)

model <- aov(value ~ factor(Dogs) + factor(Weeks) + Treat, data = BR_LatSq)

anova_table <- summary(model)[[1]]
anova_table <- cbind(Source = rownames(anova_table), anova_table)
rownames(anova_table) <- NULL

anova_table$Source <- gsub("factor\\(Dogs\\)", "Dogs", anova_table$Source)
anova_table$Source <- gsub("factor\\(Weeks\\)", "Weeks", anova_table$Source)

```

```
kable(anova_table,
      format = "latex",
      booktabs = TRUE,
      caption = "Latin square ANOVA for the BR LatSq dataset",
      digits = 3) %>%
  kable_styling(latex_options = "hold_position")
```

Section 4: R code that generates the main and interaction effects up to three-way between the factors A, B, and C in Table 5.

```
library(knitr)
library(kableExtra)

model <- lm(y ~ A * B * C, data = wtlossdat)
coefs <- coef(model)

effects <- 2 * coefs[-1]

effects_df <- data.frame(
  Effect = names(effects),
  Estimate = as.numeric(effects)
)

effects_df$Effect <- gsub(":", " x ", effects_df$Effect)

kable(effects_df,
      format = "latex",
      booktabs = TRUE,
      caption = "Estimated main and interaction effects for factors A, B, and C",
      digits = 3) %>%
  kable_styling(latex_options = "hold_position")
```

Section 4: R code that uses a linear model to find the estimated slopes of all main and interactions effects in Table 6.

```
library(knitr)
library(kableExtra)

model <- lm(y ~ A * B * C, data = wtlossdat)

coef_table <- summary(model)$coefficients
coef_df <- data.frame(
  Term = rownames(coef_table),
  coef_table
)
rownames(coef_df) <- NULL

coef_df$Term <- gsub(":", " x ", coef_df$Term)

kable(coef_df,
```

```

format = "latex",
booktabs = TRUE,
caption = "Linear model coefficients for main and interaction effects",
digits = 3) %>%
kable_styling(latex_options = "hold_position")

```

Appendix C

Section 1: R code on calculations that test the variances of treatment A versus treatment B to see if they are equal in variance.

```
var.test(fert ~ trt, data = fertdat)
```

```

##
## F test to compare two variances
##
## data:  fert by trt
## F = 0.30382, num df = 5, denom df = 5, p-value = 0.217
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.04251398 2.17122200
## sample estimates:
## ratio of variances
##          0.3038212

```

Section 1: R code that calculates the two-sample t-test using Welch's method, and the confidence interval for the `fertdat` dataset.

```
t.test(fert ~ trt, data = fertdat)
```

```

##
## Welch Two Sample t-test
##
## data:  fert by trt
## t = 0.52107, df = 7.7815, p-value = 0.6168
## alternative hypothesis: true difference in means between group A and group B is not equal to 0
## 95 percent confidence interval:
## -5.860269  9.260269
## sample estimates:
## mean in group A mean in group B
##      22.18333      20.48333

```

Section 2: R code that calculates the estimated true effect size between the two treatments in the `fertdat` dataset.

```
with(fertdat, mean(fert[trt == "A"]) - mean(fert[trt == "B"]))
```

```
## [1] 1.7
```

Section 2: R code that conducts a two-sample, two-sided t-test to find the number of samples required to hit 80% power for the **fertdat** dataset.

```
power.t.test(delta = 1.7,
             sd = sd(fertdat$fert),
             sig.level = 0.05,
             power = 0.80,
             type = "two.sample",
             alternative = "two.sided")
```

```
##
##      Two-sample t test power calculation
##
##              n = 162.9243
##              delta = 1.7
##              sd = 5.460492
##              sig.level = 0.05
##              power = 0.8
##      alternative = two.sided
##
## NOTE: n is number in *each* group
```

Section 2: R code that simulates the **fertdat** dataset, but with $n = 163$ data points for each treatment to check whether $n = 163$ achieves 80% power.

```
set.seed(123)
delta <- 1.7

sd <- sd(fertdat$fert)
sd_est <- sd(fertdat$fert)
simulate_power <- function(n, delta, sd, nsim = 5000) { pvals <- replicate(nsim, {
  groupA <- rnorm(n, mean = delta, sd = sd)
  groupB <- rnorm(n, mean = 0, sd = sd)
  t.test(groupA, groupB)$p.value })
mean(pvals < 0.05) }
n <- 163
simulate_power(n, delta, sd_est)
```

```
## [1] 0.8024
```

Section 3A: R code that conducts Tukey's post-hoc comparisons using Tukey's method.

```
TukeyHSD(anova_model)
```

```
##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = pain ~ trt, data = painstudy)
##
## $trt
```

##		diff	lwr	upr	p adj
##	B-A	1.12	0.4849214	1.755079	0.0001494
##	C-A	-2.56	-3.1950786	-1.924921	0.0000000
##	C-B	-3.68	-4.3150786	-3.044921	0.0000000